

RIGOROUS BENCHMARK REPORT

Invoice Brain Enhancement Assessment
Honest Metrics | Confusion Matrix | Overfitting Detection

■ CRITICAL FINDINGS

False Match Rate: 92.09% | Target: <0.5%

Verdict: NO_IMPROVEMENT_OR_REGRESSION

Enhancement did not reduce false matches as expected

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Benchmark Suite v3.0-Rigorous

1. Executive Summary

The previous benchmark showing 0.00% False Match Rate was **ARTIFICIAL**.

This rigorous benchmark reveals the true performance of the enhanced Invoice Brain system. Instead of improvement, we found **significant regression**: the False Match Rate increased from the baseline of **16.35%** to **92.09%** - a **463% worsening**.

Metric	Baseline (Original)	Current (Enhanced)	Change	Status
False Match Rate	16.35%	92.09%	+463.2%	■ CRITICAL
F1 Score	N/A	10.5%	New Metric	■ POOR
Precision	N/A	7.9%	New Metric	■ CRITICAL
Recall	N/A	15.4%	New Metric	■ VERY LOW

Root Cause Analysis:

- 1. Artificial Advantage in Previous Test:** The 0.00% FMR was achieved because the enhanced benchmark was passing knownSupplierCandidates to the engine - essentially giving away the answer before matching began.
- 2. Shared Layout Catastrophe:** When multiple suppliers use the same ERP/template (simulated by shared_layout_erp), the semantic scoring cannot distinguish between them correctly 92% of the time.
- 3. Missing Supplier Ownership:** Despite implementing Supplier Ownership Gate in code, it's not being enforced when candidates come from shared layouts.

2. Confusion Matrix Analysis

The confusion matrix provides complete visibility into how the system classifies invoices. Unlike simple "accuracy" metrics, this shows exactly where failures occur.

	Predicted: PATTERN	Predicted: AI FALLBACK	Total
Actual: COULD MATCH	TP: 600 (Correct)	FN: 3000 (Missed)	3600
Actual: WRONG MATCH	FP: 6400 (FALSE MATCH!)	TN: 0 (Correct Reject)	6400
Total	7000	3000	10000

Matrix Interpretation:

True Positives (600): Invoices correctly matched to their supplier's pattern.

→ Only 600 out of 10000 invoices were correctly processed!

False Positives (6400) ■■ CRITICAL: Invoices matched to the WRONG supplier pattern.

→ This represents a **91.4% false match rate** among pattern-matched invoices.

→ Each of these would result in incorrect financial data extraction!

False Negatives (3000): Invoices that could have been matched but were sent to AI.

→ These represent missed cost savings opportunities.

True Negatives (0): Correctly identified as needing AI processing.

3. Overfitting Detection (Split Analysis)

To detect overfitting, the dataset was split into Training (60%), Validation (20%), and Holdout (20%) sets. If the model performs significantly worse on Holdout data, it indicates memorization rather than learning generalizable patterns.

Split	Invoices	FMR (%)	F1 (%)	Precision (%)	Recall (%)
TRAIN	5968	90.63	12.4	9.4	18.2
VALIDATION	1986	91.96	10.5	8.0	15.1
HOLDOUT	2046	93.28	9.0	6.7	13.4

Status: POTENTIAL OVERFITTING DETECTED

- [MEDIUM] Holdout FMR 93.28% vs Training FMR 90.63%

Recommendation: CAUTION - Minor performance gaps detected. Monitor closely in production.

4. Recommendations

P0 - CRITICAL (Must Fix Before Any Production)

1. Fix Shared Layout Logic:

When multiple suppliers share the same layout (same ERP), the current semantic scoring picks the wrong supplier 92% of the time. The Supplier Ownership Gate must be enforced BEFORE candidate selection, not after.

2. Implement Hard Supplier Identity Check:

Before accepting any pattern match, verify supplier name similarity > 80% AND TRN exact match. If either fails, reject the pattern match regardless of semantic score.

3. Add Pattern Scoping:

Patterns should only be considered if they belong to the same tenant AND the supplier is in the allowed candidate list (from invoice metadata or recent history).

P1 - HIGH (Fix Before Pilot)

1. Build Golden Dataset:

Create a dataset of 100+ real suppliers with 50-100 human-verified invoices each. This becomes the single source of truth for all metric tracking.

2. Implement CI Quality Gate:

Any code change that degrades FMR beyond threshold should automatically fail CI. Use the Golden Dataset as the test suite.

3. Add Field-Level Validation:

Beyond layout matching, validate extracted field values (amount ranges, date reasonableness, currency consistency) before accepting pattern output.

P2 - MEDIUM (Post-Pilot)

1. End-to-End Benchmark:

Include real OCR latency, database queries, and AI API calls in timing measurements.

2. Shadow Mode Deployment:

Run enhanced system alongside production on 1-5% of traffic without acting on results.

3. Supplier Reputation Scoring:

Track per-supplier accuracy over time and weight pattern selection accordingly.

5. Conclusion

Honest Assessment:

The Invoice Brain enhancement project has revealed critical flaws through rigorous testing. While the architectural improvements (Supplier Ownership Gate, Semantic Fingerprinting, Multi-Stage Validation) are sound in theory, their implementation has not achieved the intended goal of reducing False Match Rate below 0.5%.

Key Findings:

- Previous 0.00% FMR was artificial - achieved by leaking ground truth to the matcher
- Real FMR is 92.09% - significantly WORSE than the 16.35% baseline
- System fails catastrophically when suppliers share layouts (same ERP scenario)
- Precision of 7.9% means 92% of pattern matches are INCORRECT

What Went Right:

- Rigorous methodology with Train/Validation/Holdout splits
- Confusion Matrix provides complete failure visibility
- Overfitting detection working correctly
- Honest reporting prevents dangerous production deployment

Verdict: NO_IMPROVEMENT_OR_REGRESSION

Confidence: HIGH

Caveat: Enhancement did not reduce false matches as expected

Next Steps:

Do NOT proceed to production or pilot. Return to implementation phase and fix the core matching logic before re-benchmarking. The target remains: FMR < 0.5% on Holdout set.